

Experiment No: 7

Title:

Perform DOM traversal and updates; develop a to-do list app with add, edit, and delete features.

Software/Tools Required:

- JavaScript
- HTML
- CSS
- Code Editor

Theory:

DOM Traversal and Manipulation in JavaScript

The Document Object Model (DOM) is a programming interface that represents an HTML document as a tree structure. Each element, attribute, and piece of text in the webpage is represented as a node in this tree. JavaScript can access, modify, add, or remove these nodes dynamically, allowing developers to create interactive web applications.

DOM traversal refers to navigating through different nodes of the document, while DOM manipulation involves updating the content, structure, and style of web page elements.

1. Document Object Model (DOM)

When a web page loads, the browser creates a DOM representation of the HTML document. JavaScript can interact with this structure to access and modify webpage content dynamically.

2. DOM Traversal

DOM traversal is the process of navigating between elements in the DOM tree.

Common traversal properties include:

a) `parentElement`

Accesses the parent element of a node.

b) `children`

Returns all child elements of a node.

c) firstElementChild

Returns the first child element.

d) lastElementChild

Returns the last child element.

e) nextElementSibling and previousElementSibling

Access adjacent sibling elements.

3. DOM Manipulation

JavaScript can dynamically modify webpage content and structure.

a) Changing Content

Example:

```
document.getElementById("msg").innerHTML = "Welcome";
```

b) Changing Styles

Example:

```
document.getElementById("msg").style.color = "blue";
```

c) Creating Elements

Example:

```
let item = document.createElement("li");
```

d) Appending Elements

Example:

```
list.appendChild(item);
```

e) Removing Elements

Example:

```
item.remove();
```

Case Study

Create the given HTML Registration Form and use JavaScript to access and manipulate its form elements. Validate user inputs such as Firstname, Email, and Password, including checking for empty fields and valid email format. Implement various form events such as submit, change, and focus to make the form interactive and user-friendly.

To-Do List

Add your task here...

Add

☐ JS NOTES TAKING Edit ×

☐ CN LAB Edit ×

Registration Form

*Mandatory to fill

Firstname: *

Lastname:

Birthday:

Username: *

E-mail:

Website:

Password: *

Re-password: *

☒ I agree to the terms & conditions.

Register

Registration Successful!

Conclusion:

The Document Object Model (DOM) enables JavaScript to access and manipulate webpage elements dynamically. DOM traversal techniques help navigate through the document structure, while DOM manipulation methods allow content creation, modification, and deletion. A To-Do List application effectively demonstrates these concepts by enabling users to add, edit, and delete tasks dynamically, making web applications more interactive and user-friendly.